

Chapter#4

Introduction to web design and HTML

Website:

A website is a collection of related web pages that are connected and available on the internet under one domain name.

Example:

google.com, facebook.com, amazon.com

Webpage:

A single document on a website that displays: Text, Images, Videos, Links etc.

Example:

A product page on Amazon.

A website can contain:

- Text, Images, Audio, Video, Databases, Hyperlinks etc.

A website can be owned by:

- A company
- The government
- An educational institute
- An individual person
- A hospital

Advantages/Benefits of website:

- Easy to communicate
- Gives quick access to info
- Saves paper and time
- Helps in marketing and selling
- Reaches more people
- Good for publicity and ads
- Works from anywhere, anytime
- 24/7 hours services

Disadvantages of a Website:

- Info may be wrong or fake
- Needs internet to use
- Site may not open if server is down
- Can be hacked
- Privacy can be at risk
- Costly to build and maintain
- Needs regular updates
- Slow speed can cause problems
- May not work well on all devices

Birth of the Internet & Web

- Need to transfer files led to the creation of the **World Wide Web**.
- Invented by **Tim Berners-Lee** in **1989** at CERN, Switzerland.
- Each computer is uniquely identified by its IP address(Public IP address)
- An **IP address (Internet Protocol address)** is a unique numerical identifier assigned to a device on a network.
- **Example**
 - 92.168.4.3
 - IPv4 is **32 bit** address

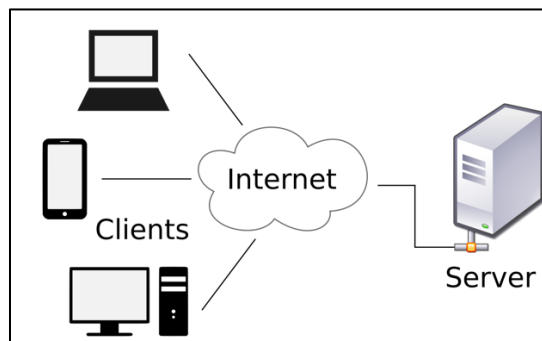
Key Concepts by Tim Berners-Lee

- **Hypertext**: Linked text on websites.
- **Hyperlink**: Address of the hypertext.
- **Hypermedia**: Media files like photo, video, audio on the web.
- Created **Web Browser** to view web documents.
- Enabled easy navigation with just a click.

Web client:

A web client is a computer connected to the Internet that allows users to browse websites from servers.

Example: Mobile, Tab, Laptop, PC etc.



Web server:

A web server is a computer connected to the Internet that hosts (stores) websites

Upload:

Transferring information from a client's computer to a server.

Download:

Transferring information from a server to a client's computer.

Ans.

Sample MCQ

1. Image, video, audio on internet called--
a) hyper link b) hyper text c) hyper media d) hyper voice 1.c
2. The purpose of a web server is to _____
a) host websites b) store files in browser c) access websites from client d) create websites 2.a
3. Websites can contains ----
i. Database
ii. Hyper link
iii. Video
Which is correct?
a) i & ii b) i & iii c) ii & iii d) i, ii & iii 3.d
4. How many bits remain in the IPv4 address?
a) 2 b) 8 c) 16 d) 32 4.d
5. Shuvo went to a cybercafé and filled up an electronic application form for college admission. Sitting there he received an SMS of confirmation of the successful submission of the form. What type of IT service did Shuvo receive?
a) Web hosting b) E-Commerce c) Electronic mail d) Websites browsing 5.d
6. Who is the inventor of WWW?
a) Tim Berners Lee b) Mark Zuckerberg c) Steve Jobs d) Charles Babbage 6.a
7. What is the term for the clickable elements on a web page that lead to other locations online?
a) Icons b) Images c) Hyperlinks d) Buttons 7.c
8. Transferring information from client to server is called-----
a) upload b) hyperlink c) download d) client 8.a

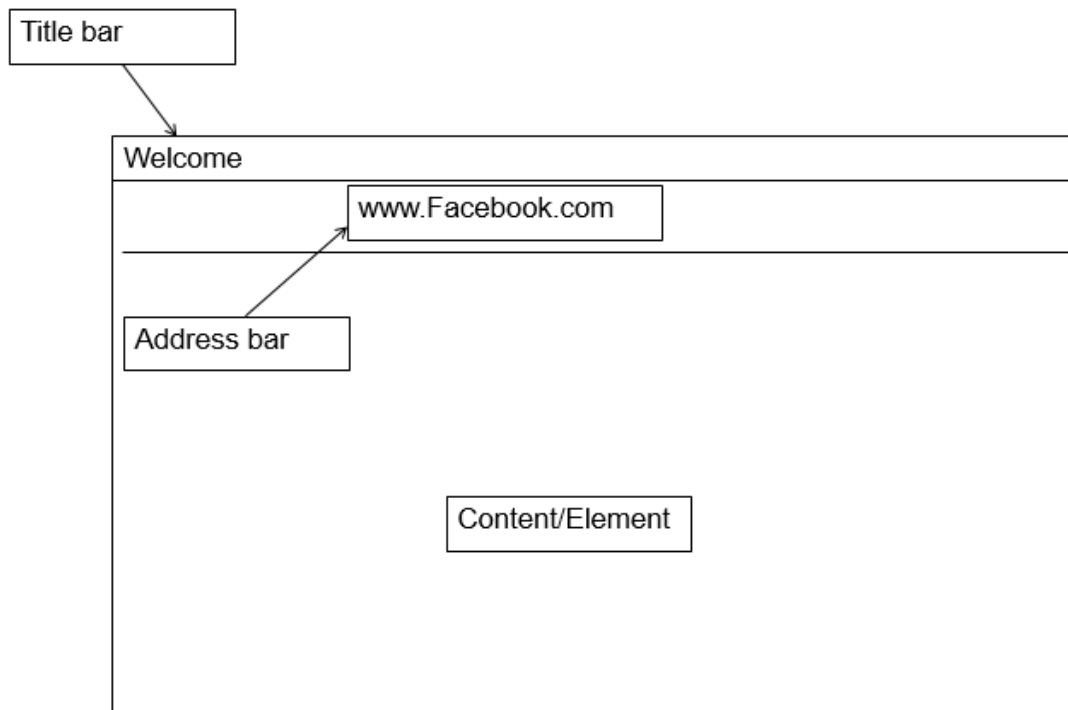
Browser:

A web browser is a software that helps you visit websites.

Example:

- Mozilla Firefox
- Google Chrome
- Safari
- Internet Explorer
- Microsoft Edge

Internet connection and web browser needed to visit/browse the websites.



Sample MCQ

Ans.

1. The primary function of a web browser is to ----
 a) Store data on your computer b) Run video games c) Access and interact with websites d) Write computer programs 1.c
2. Which software are used to display data on the website?
 a) Paint and Word b) Notepad and SQL c) chrome and Mozilla d) Norton and bKash 2.c
3. To visit a home page we must need.....
 i) web browser
 ii) search Engine
 iii) Internet
 Which is correct?
 a) i & ii b) i & iii c) ii & iii d) i, ii & iii 3.b
4. Someone can visits a website by----
 i) Google chrome
 ii) MySQL
 iii) Mozilla
 Which is correct?
 a) i & ii b) i & iii c) ii & iii d) i, ii & iii 4.b
5. Which part of the browser is located at the top?
 a) address bar b) title bar c) close button d) open 5.b
6. What is the address bar in a web browser used for?
 a) Entering Uniform Resource Locators b) Managing bookmarks c) Displaying search results d) Viewing browsing history 6.a

Network protocol:

A set of rules that provide communication in a network called protocol.

Examples:

HTTP= Hyper Text Transfer Protocol (used for browsing)

FTP= File Transfer Protocol

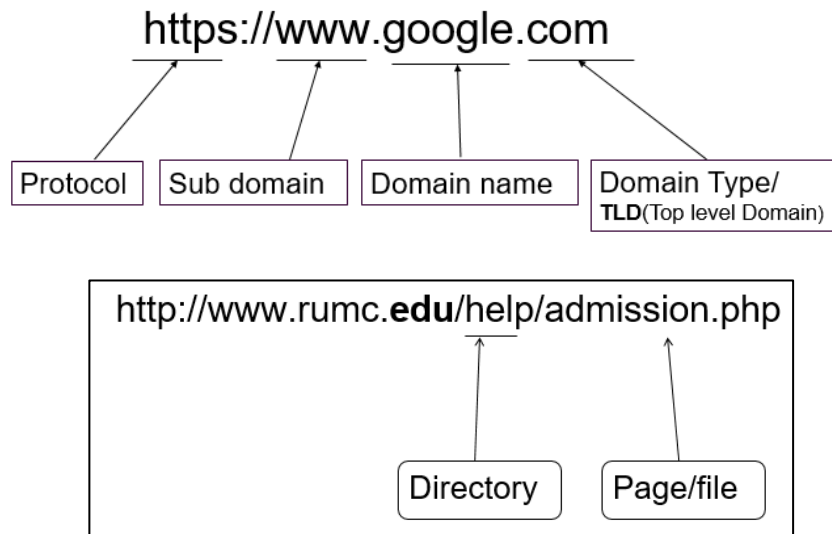
TCP/IP = Transmission Control Protocol/Internet Protocol

URL (Uniform Resource Locator):

A URL is the address of a webpage on the Internet.

Example:

<https://www.google.com>



Ans.

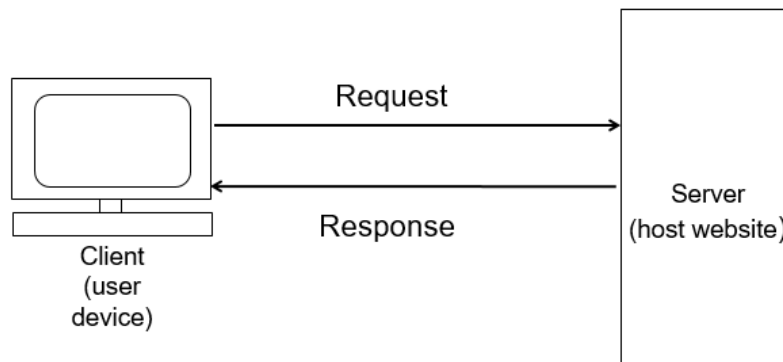
Sample MCQ

- The full form of URL is -----
 a) Uniform Resource Locator b) Uniform Research Locator c) Uniform Reaching Locator d) Unit Resource Locator 1.a
- Which is NOT the task of HTTP?
 A. connection between browser & server B. transmit the client's request to the server C. change and enhance data while reaching to client D. stop transmission when the transmission is done 2.c
- Usually, the address used on Internet---
 i) IP address ii) Text-based DNS iii) Local Address
 Which is correct?
 a) i & ii b) i & iii c) ii & iii d) i, ii & iii 3.a
- What is HTTP?
 a) Hyper Text Transfer Protocol b) Hyper Test Transfer Protocol c) Higher Test Transfer Protocol d) Hyper Text Transection Protocol 4.a
- <http://www.yahoo.com> what is the last part?
 a) protocol b) domain name c) file type d) domain type 5.d

6. The unique address of web site....
- a) IP address b) URL c) HTTP d) domain type 6.b
7. HTTP is a browsing _____
- a) website b) search engine c) web portal d) protocol 7.d
8. http://www.rajshahiboard.gov- what is the top-level domain here?
- a) http b)www c) gov d)board 8.c
9. What is called the rule of receiving and sending data in a computer network?
- a. URL b. Protocol c. Browser d. IP address 9.b
- 10 Which represents the domain name?
- a) Internet b) URL c) Yahoo d) google 10.b

Client –Server relationship

- **Client:** Sends user input (Request).
- **Server:** Processes input and sends back result (Response).



Example:

Creating an account on a website involves:

1. Client sends form data.
2. Server checks and processes.
3. Sends success/error message.

Programming for Web:

Server-side languages: PHP, Python, Ruby, Java.

Client-side technologies:

- **HTML:** Structures content (not a programming language).
- **CSS:** Styles the website (colors, layout).
- **JavaScript:** Handles client-side actions and interactivity.

HTML is used to create the structure of a webpage (like text, images, buttons), while CSS is used to style it (like colors, fonts, layout).

Using them together makes the webpage both **organized** and **attractive**.

Ans.

Sample MCQ

- | | | | | | |
|---|--------------------------------------|---------------------------------------|----------------------------------|---------------------------------------|-----|
| 1. Examples of server side software are--- | a) Python, Ruby | b) HTML,CSS | c) C/C++ | d) JavaScript, R | 1.a |
| 2. What is the full form of CSS? | a) Case Style Sheet | b) Computer Style Sheet | c) Cascading Style Shift | d) Cascading Style Sheet | 2.d |
| 3. Relation between web client and server--
i) client send request
ii) server checks the validity of request
iii) server response with feedback
Which is correct? | a) i & ii | b) i & iii | c) ii & iii | d) i, ii & iii | 3.d |
| 4. Which type of website allows client-side programing? | a) static | b) dynamic | c) fixed | d) structure | 4.b |
| 5. What is the full form of HTML? | a) Hi Text Markup Language | b) Hyper Text Markup Language | c) Hyper Text Makeup Language | d) Hyper Text Markup Level | 5.b |
| 6. What is sent by client to server? | a) acknowledge | b) response | c) request | d) feedback | 6.c |
| 7. To make a website look visually appealing, what is commonly used in conjunction with HTML? | a) PHP | b) Python | c) Ruby | d) CSS | 7.d |
| 8. What are tags used for in a markup language? | a) To define variables and functions | b) To insert comments and annotations | c) To structure and display data | d) To perform mathematical operations | 8.c |

Types of websites (based on runtime changes)

- 1) Static Websites
- 2) Dynamic Websites

1) Static Websites

Static Websites are simple sites built using HTML and CSS with fixed pages that cannot take user input or use a database.

Example:

Company brochure websites, small business info sites, or documentation pages built without a database.

2) Dynamic Websites:

Dynamic Websites are interactive sites that use server-side languages like PHP or Python with databases (e.g., MySQL, SQL Server) to generate new pages and take user input.

Example:

YouTube, Amazon, Twitter, Gmail, and online banking websites

Static vs Dynamic Websites

Static Websites	Dynamic Websites
1) Can't take input from user	1) Can take input from user
2) No database used	2) Database used
3) Use HTML and CSS	3) Use PHP, Python, etc. with database (MySQL/SQL Server)
4) Fixed number of pages	4) New pages can be generated
5) Hard to update	5) Easy to update
6) One-way communication	6) Two-way communication
7) Simple structure & easy to develop	7) Complex structure & hard to develop
8) Faster loading time	8) Slower loading time
9) Less expensive to build	9) Expensive
10) User login system is not possible	10) User login system is possible
11) Company brochure websites, small business info sites, or documentation pages built without a database.	11) YouTube, Amazon, Twitter, Gmail, and online banking websites

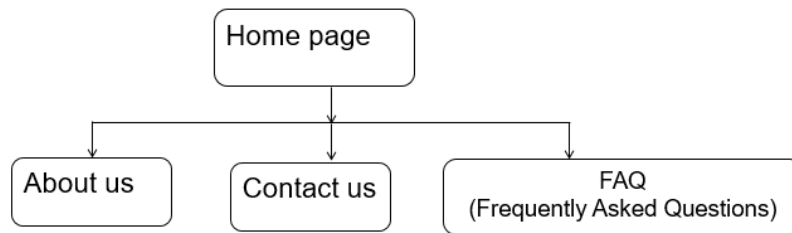
Sample MCQ

- | | Ans. |
|--|------|
| 1. Based on the change of runtime how many types of websites?
a) 2 b) 3 c) 4 d) 5 | 1.a |
| 2. Websites built with HTML and CSS are most likely ---
a) Dynamic websites b) Static websites c) Mobile-friendly websites d) CMS websites | 2.b |
| 3. Developing a dynamic website requires -----
i. PHP ii. Python iii. MySQL
Which of the following is correct?
a) i & ii b) i & iii c) ii & iii d) i, ii & iii | 3.d |

4. Which languages are used for creating static websites?
- a) PHP & MySQL b) ASP & Python c) CSS & HTML d) JSP & XML 4.c
5. Joy and Shaheed developed a website using HTML where each page can be visited from any page and the website can be loaded faster. Later because of some problems they use MySQL and PHP and some tools to change the website. Their decision may lead to_____
- i) easy change the content
ii) can add a database
iii) increase the security of the website
- Which of the following is correct?
- a) i & ii b) i & iii c) ii & iii d) i, ii & iii 5.d
6. For developing a web-based database you will need :
- a) MS Access b) SQL Server c) Debase d) FoxPro 6.b
7. Static Website —
- i) Faster loading time
ii) Difficult to update content
iii) Developed mainly using HTML
- a) i & ii b) i & iii c) ii & iii d) i, ii & iii 7.d

Sample CQ
Students at the higher secondary level of a college were divided into two groups and instructed to develop a website. The first group developed their website using HTML and CSS, while the second group used CSS, MySQL, PHP, etc., to develop theirs. The judges selected the second website. Q) Explain why the judges selected the second website.
The teacher discussed two types of websites in his lecture, explaining that one is interactive while the other is not. Q) Based on the teacher's discussion in the passage, which of the two types of websites do you think is more suitable for use?
By using the website of 'KHA' College, one can fill out the admission form, input examination marks, and guardians can exchange their views and opinions related to education. Q) What type of website is mentioned in the above stem?
The website of Shila's college provides two-way communication between the server and the browser. Q) Describe the type of website mentioned in the above stem.
Two friends, Ela and Anjum, developed a website. Though the website can be browsed quickly, the information cannot be updated regularly. That's why they decided to rebuild the website using tools like PHP, MySQL, etc. Q) Analyze the logic behind the decision mentioned in the stem.
Risha wanted to develop a website, so she consulted a web developer. The web developer suggested two methods: one is cheap but hard to update regularly, and the other is costly but easier to update. Q) What type of websites are mentioned in the stem?

Structure a website

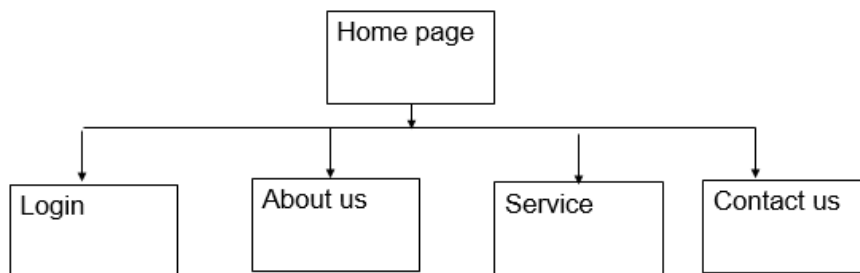


A website has one or more web pages. The first page is called the **Homepage**. Websites have different pages based on their purpose. For example, audio/video sites have pages to play videos. Most sites have a **profile page** for users. **Blog sites** have a page for each blog post. Some common pages are **Contact Us**, **About Us**, and **FAQ**.

Types of Websites Based on Structure

- A) Hierarchical/Tree structure
- B) Linear/Sequential structure
- C) Network/Relational structure
- D) Hybrid structure

A) Hierarchical/Tree structure:

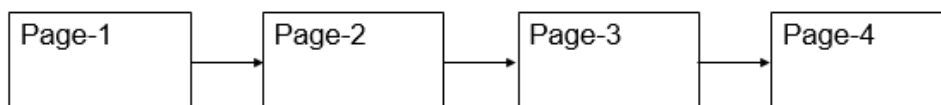


A tree-like structure where the homepage acts as the root, leading to main categories and subcategories.

Example:

Common in corporate and e-commerce sites.

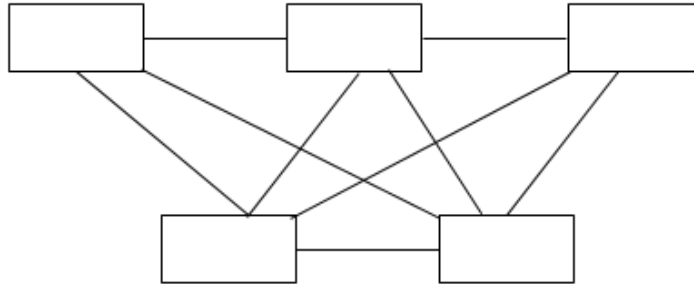
B) Linear/Sequential structure



Pages are arranged in a step-by-step sequence, guiding users through a predefined order.

Example:

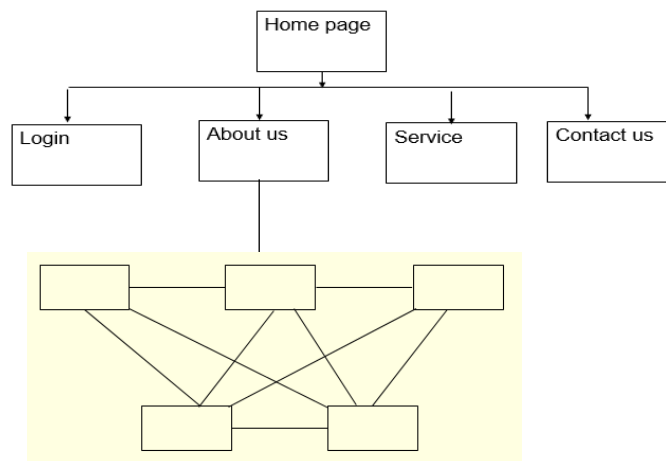
Used in tutorials, online courses, and presentations.

C) Network/Relational structure

Pages are interconnected in a web-like manner, allowing users to navigate freely between related content.

Example:

Found in wikis and research websites.

D) Hybrid structure:

A combination of multiple structures, typically mixing hierarchical and network models for better navigation.

Example: Common in complex websites like news portals and educational platforms.

Ans.

Sample MCQ

- | | | | | |
|----|--|---------------------|----------------------|---------------------|
| 1. | Which model of the website consists of 2 or more models? | | | |
| | a) sequential | b) network | c) hybrid | d) relational |
| | | | | 1.c |
| 2. | Which structure is commonly used for online tutorials? | | | |
| | A) Hierarchical Structure | B) Linear Structure | C) Network Structure | D) Hybrid Structure |
| | | | | 2.b |

3. Usually an website consists of—

- i) FAQ
- ii) home page
- iii) contact us

Which of the following is correct?

a) i & ii

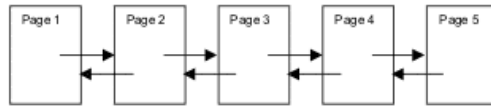
b) i & iii

c) ii & iii

d) i, ii & iii

3.d

4. The figure indicates _____structure of website



a) relational

b) network

c) hybrid

d) sequential

4.d

5. The figure indicates _____structure of website



a) relational

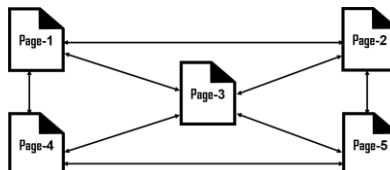
b) hierarchical

c) hybrid

d) sequential

5.b

6. The figure indicates _____structure of website



a) relational

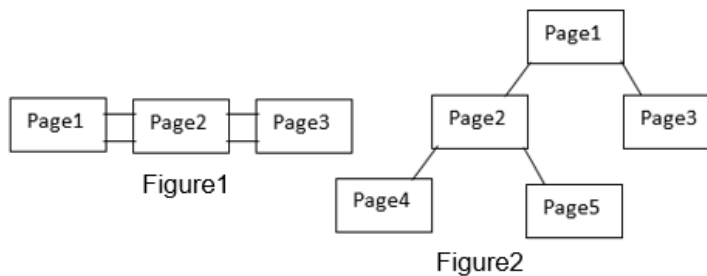
b) hierarchical

c) hybrid

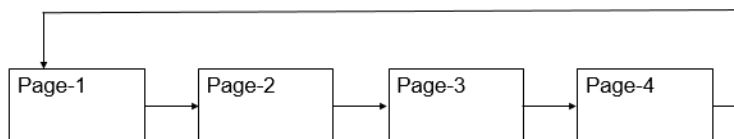
d) sequential

6.a

Sample CQ



Q) What types of website structures are shown in Figure 1 and Figure 2? Which one is better?

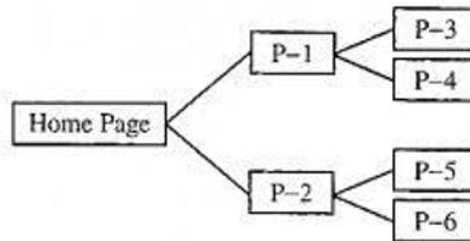


Q) What type of website structure is shown in this figure?

A university website has a homepage that contains links to different faculties. Each faculty page further contains links to departments, and each department page provides links to faculty members and course details.

Q) What type of website structure is used in this scenario?

Figure 1



Q) What type of website structure is shown in this figure?

An online training program offers a step-by-step learning experience. Users must complete each module before proceeding to the next. Navigation is strictly sequential, ensuring learners follow a predefined learning path.

Q) What type of website structure does this training program follow?

A news portal organizes its content in a main menu with sections like "Politics," "Sports," and "Entertainment." However, within articles, there are also links to related news stories, allowing readers to explore topics non-linearly.

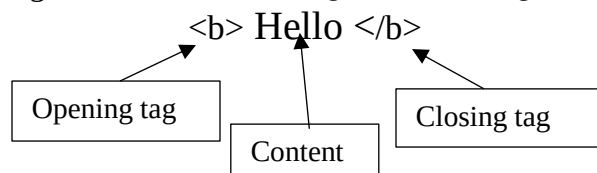
Q) What type of structure does this news portal primarily use?

Basic idea of HTML

- HTML is used to create webpages.
- File extension must be .html (e.g., mypage.html).
- Open with a web browser → shows output.
- Edit using **text editors** like **Notepad**, **Notepad++** and **Sublime Text**.

HTML Tags

Tags are like brackets: <tagname> and </tagname>.



Content is written **between** opening and closing tags.

Example:

```
<b>Hello</b>
```

A simple HTML file

```
<!DOC Type html>
<html>
  <head>
    - - - - -
  </head>

  <body>
    - - - - -
  </body>
</html>
```

Formatting tags:

HTML formatting tags are used to change the appearance of text on a webpage.

Example of some common formatting tags:

tag	Purposes
	Used to display bold text
<i>	Used to display italic text
<u>	Used to display underlined text

 tag:

Used to display bold text

Example:

```
<!DOC Type HTML>
<html>
  <head> </head>
  <body>
    <b> Bangladesh </b>
  </body>
</html>
```

Note: tag can also serve the same purposes

Result:

Bangladesh

<i> tag:

Used to display italic text

Example:

```
<!DOC Type HTML>
<html>
  <head> </head>
  <body>
    <i> Uttara </i>
  </body>
</html>
```

Result:

Uttara

Note: tag can also serve the same purposes emphasize (italic)

<u> tag:

Used to display italic text

Example:

```
<!DOC Type HTML>
<html>
  <head> </head>
  <body>
    <u> Dhaka </u>
  </body>
</html>
```

Result

Dhaka

Note: <ins> tag can also serve the same purposes

Different tags can be applied to the same text:

```
<!DOC Type HTML>
<html>
  <head> </head>
  <body>
    <b><u><i>Bangladesh</i></u></b>
  </body>
</html>
```

Result:

Bangladesh

Notes: The sequence is necessary because HTML tags must be properly nested.

Open tags from outside → inside

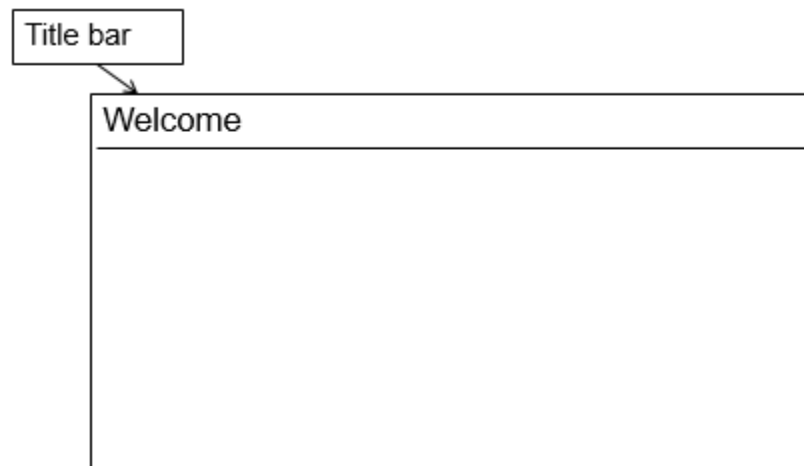
Close tags from inside → outside

Displays the text in the browser's title bar:

The title tag should be placed inside the <head> section

```
<!DOC Type HTML>
<html>
  <head> <title> Welcome </title>
</head>
  <body>
</body>
</html>
```

Result:



Remarks/comments:

In HTML, **remarks/comments** are notes in the code that are not displayed on the web page. They are written between <!-- and --> and are used to explain code, leave reminders, or disable parts of the code temporarily.

Example:

```
<!-- DOC Type HTML -->
```

Ans.

Sample MCQ

1. The extension of an HTML file is _____

a) .http

b) .html

c) .txt

d) .c

1.b

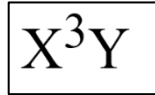
2. Which is a HTML editor?
a) Notepad b) Paint c) Mozilla d) MS Word 2.a
3. The HTML editor can be—
i) Notepad++
ii) Sublime Text
iii) chrome
Which of the following is correct?
a) i & ii b) i & iii c) ii & iii d) i, ii & iii 3.a
4. Feature of html---
i) case sensitive
ii) supported by all browsers
iii) need to use tag
Which of the following is correct?
a) i & ii b) i & iii c) ii & iii d) i, ii & iii 4.c
5. Formatting tags are---
a) ,<th>,<src> b) <table>,,<td> c) <a>,,<href> d) ,<i>,<u> 5.d
6. Which tag is used to make text bold?
a) <a> b) c) <i> d) <u> 6.b
7. Which tag is used to make text underlined?
a) <a> b) c) <i> d) <u> 7.d
8. Which tag is used to make text italic?
a) <a> b) c) <i> d) <u> 8.c
9. Which is the correct example of writing tag?
a) <i> abc<i> b) </i> abc<i> c) <i> abc</i> d) <i> abc</i> 9.d
10. Which are basic tags?
a) ,, b) <a><table> c) <html><head><body> d) ,<p>,<div> 10.c
11. Which symbols are used for a tag?
a) { } b) <> c) [] d) \$\$ 11.b
12. <title> tag is used under ____
a) <body> b) <h1> c) <p> d) <head> 12.d

<sup> tag:

Displays text as *superscript*.

```
<!DOC Type HTML>
<html>
  <head>  </head>
  <body>
    X <sup>3 </sup>Y
  </body>
</html>
```

Result:

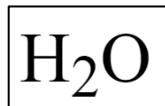


<sub> tag:

Displays text as *subscript*.

```
<!DOC Type HTML>
<html>
  <head>  </head>
  <body>
    H <sub>2 </sub> O
  </body>
</html>
```

Result:



Sample MCQ

Ans.

- What is the output of the following tag?
A₂B³
a) A₂B₃ b) A²B³ c) A²B₃ d) A₂B³ 1.d
- What is the correct tag for seeing P²Q₃ as output?
a) P²Q₃ b) P</sub>2<sub>Q₃ c) P²Q₃ d) P²Q₃ 2.a
- What is the correct tag for seeing the following output?
$$X^{2^N}$$

a) X²N b) X^{2^N c) X}2^N d) X²N^N 3.b
- What is the meaning of <sub> in HTML?
a) Superscript text b) Subscript text c) Small text d) strong text 4.b

Heading tags : <h1>, <h2>, <h3>, <h4>, <h5>, <h6>

Used to define headings:

<h1> tag:

```

<!DOC Type HTML>

<html>

  <head>  </head>

  <body>

    <h1>Today's Headline</h1>

  </body>

</html>

```

Result:

Today's Headline

Example of Headline tags:

```

<!DOC Type HTML>
<html>
  <head>  </head>
  <body>
    <h1> Headline 1</h1>
    <h2> Headline 2</h2>
    <h3> Headline 3 </h3>
    <h4> Headline 4 </h4>
    <h5> Headline 5 </h5>
    <h6> Headline 6 </h6>
  </body>
</html>

```

Headline 1

Headline 2

Headline 3

Headline 4

Headline 5

Headline 6

Ans.

Sample MCQ

1. What is the smallest heading tag in HTML?

a) <h1> b) <h3> c) <h6> d) <h7> 1.c

2. What is the highest heading tag in HTML?

a) <h1> b) <h3> c) <h6> d) <h7> 2.a

3. `<h2> ABC </h2>` the output text “ABC” will be ---
 i) larger
 ii) bold
 iii) underlined
 a) i & ii b) i & iii c) ii & iii d) i, ii & iii 3.a
4. What is the valid range of heading tags in HTML?
 a) `<h0>` to `<h5>` b) `<h1>` to `<h6>` c) `<h1>` to `<h5>` d) `<h2>` to `<h7>` 4.b
- 5.

```
<html>
<body>
<h1> ICT </h1>
</body>
</html>
```

What is the output of the HTML code?

- a) Text will be in larger size b) Text will shown as title c) Text will be scrolled d) hyper link will be created 5.a

** tag:**

`<del>` tag shows deleted text, usually displayed with a strikethrough.

```
<!DOC Type HTML>
<html>
  <head> </head>
  <body>
    <del> My old phone number </del>
  </body>
</html>
```

Result:

~~My old phone number~~

Note: `<strike>` tag serves similar purpose as `<del>`

Sample MCQ

Ans.

1. Which is alternate of the `<del>` tag?
 a) `<img>` b) `<strike>` c) `<div>` d) `<font>` 1.

2. What is the output of the below html tag?

`<u> This is </u> <del>RUMC</del>`

- a) ~~This is~~ RUMC b) This is ~~RUMC~~ c) This is RUMC d) ~~This is RUMC~~

2.

3. Correct tag for the output mentioned below are____

i) `< del >`

ii) `< strike >`

iii) `< div >`

~~Dhaka~~

Which of the following is correct?

a) i & ii

b) i & iii

c) ii & iii

d) i, ii & iii

3.

**
 tag:**

Used to insert a **line break** in text. It's an **empty** tag (no closing tag needed).

```
<!DOC Type HTML>
<html>
  <head> </head>
  <body>
    Hello! <br> World
  </body>
</html>
```

Result:

Hello!
World

Sample CQ

Write HTML code to produce the

Quick *Brown* fox
~~jump over~~ the lazy^{dog}
and then **it fall**
prey to a lion

output shown in the above stem.

<p> tag:

Defines a **paragraph** in HTML. Automatically adds some space (margin) before and after the paragraph.

```
<!DOC Type HTML>
<html>
  <head> </head>
  <body>
    <p>DHAKA</p><p>
    BANGLADESH</p>
  </body>
</html>
```

Result:

DHAKA
BANGLADESH

<p> tag with attribute:

```
<!DOC Type HTML>
<html>
  <head> </head>
  <body>
    <p align="center"> Dhaka </p>
  </body>
</html>
```

Result:

Dhaka

Identify the tag, attribute and value

<p align="center"> Dhaka </p>

Opening tag

Attribute

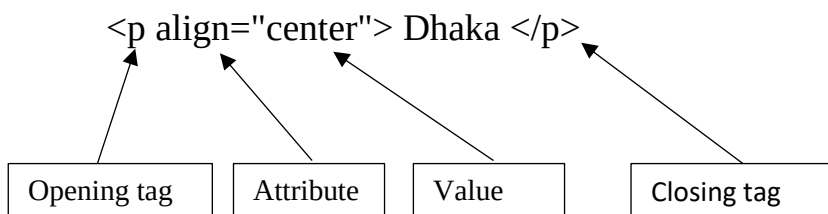
Value

Closing tag

The value "left" and "right" is also possible with align attribute.

Attribute:

An **attribute** is a modifier of an HTML element that provides extra information about that element. Attributes are always included in the opening tag and usually come in name-value pairs.

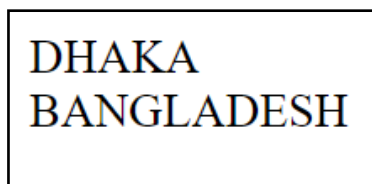


Value:

In HTML, the **value** is the **specific information assigned to an attribute** of an HTML element. It tells the browser how to apply that attribute to the element.

<div> tag:

The <div> tag is used to divide a document into separate sections.



The align attribute can also be used with the div tag.

Sample MCQ

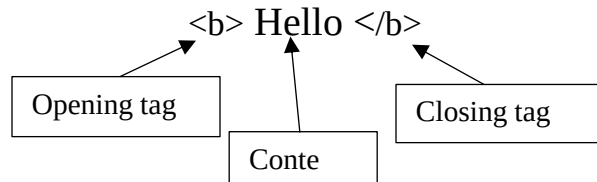
						Ans.
1.	Which tag creates more space between two lines?	a) <div>	b) 	c) <p>	d) 	1.c
2.	What is the correct syntax for aligning a <p> tag to the center using the align attribute?	a) <p align =centre>	b) <p align = "center">	c) <p align : center>	b) <p center= "align">	2.b
3.	Value of align can be: iii) top is correct?	i) left	ii) center	iii) top		3.a
	Which of the following is correct	a) i & ii	b) i & iii	c) ii & iii	d) i, ii & iii	3.a
4.	Which is the attribute value of the following HTML code?	b) <p align = "right"> ICT </p>				
		a) p	b) ICT	c) right	d) align	4.c
5.	Which can be alternate of <div> tag?	a) <a>	b) 	c) <i>	d) 	5.d

Pair tag/container tag

A pair tag has both an opening and a closing tag.

Also called double tag

Syntax:



Examples:

- `<b> </b>`
- `<i> </i>`
- `<p> </p>`
- `<u> </u>`

Non Pair tag/Empty tag:

An empty tag is an HTML tag that does not have a closing tag and cannot enclose content.

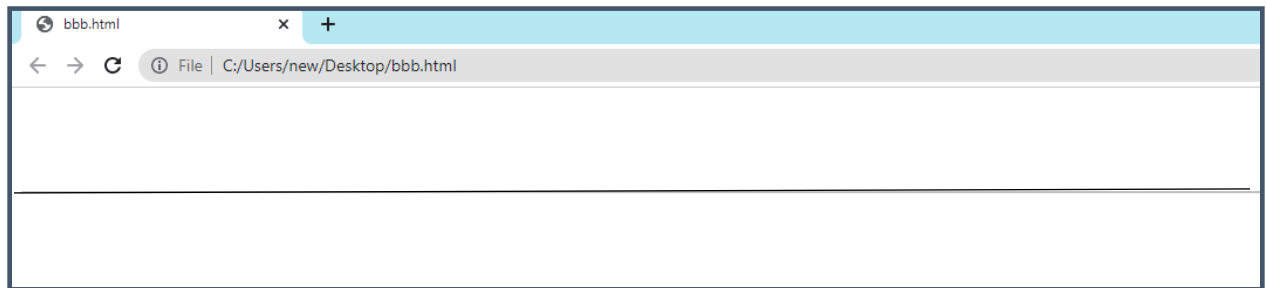
Also called self closing tag.

Example:

- `<br>`
- `<hr>`
- `<img>`

`<hr>` Tag:

Used to draw a horizontal line in an HTML page, typically for separating sections of content.



Sample MCQ

Ans.

- | | | | | |
|--|---------------------------|----------------------------|----------------------------|-----|
| 1. Which is a non-pair tag? | | | | |
| a) <code><a></code> | b) <code></code> | c) <code>
</code> | d) <code><tr></code> | 1.c |
| 2. Which tag has no closing tag? | | | | |
| a) table tag | b) empty tag | c) pair tag | d) basic tag | 2. |
| 3. Example of empty tag: i) <code></code> ii) <code><hr></code> iii) <code>
</code> Which is correct ? | | | | |
| a) i & ii | b) i & iii | c) ii & iii | d) i, ii & iii | 3.d |

4. Which one is an empty tag?
 a) <hr> b) c) <caption> d) <tr> 4.a
5. Which one is a pair tag?
 a) <hr> b) c)
 d) 5.d
6. What is the use of the
 tag?
 a) page break b) line break c) Delete page d) hyperink 6.b
7. The tag <HR> means---
 a) heading of table b) heading new c) horizontal line d) header file 7.c

Sample CQ

```
<html>
<head><title> my web</title></head>
<body>
<i>Uttara</i> <br> <b>Dhaka </b>
<br><u>Bangladesh </u><br>
H<sub>2 </sub>O
</body>
</html>
```

What is the output of the above html code?

Head line

Dhaka

Bangladesh

A⁴B₂

Write HTML code to the show the above output.

Advantages / Benefits of HTML for Creating Websites:

- Very easy to learn and use
- Free – no need to buy software
- Not case sensitive
- Works on all browsers
- Platform independent – runs on any device
- Can show text, images, audio, video
- Helps create links to other pages (hyperlinks)
- Can be used with CSS and JavaScript for better design and interactivity
- Used to build the structure of every website

Disadvantages of HTML:

- Can only create static websites
- Must write many tags manually
- No built-in security
- Must memorize many tags
- Can't connect with a database directly
- Not suitable for complex or dynamic websites
- Design and layout are limited without CSS

 tag:

An HTML tag used to change the **font face**, **size**, and **color** of text.

Example1:

```
<!DOC Type HTML>
<html>
  <head>  </head>
  <body>
    <font color="red"> Uttara </font>
  </html>
</body>
</html>
```

Example 2:

```
<!DOC Type HTML>
<html>
  <head>  </head>
  <body>
    <font size="7" color="Blue"> Uttara </font>
  </body>
</html>
```

Example 3:

```
<!DOC Type HTML>
<html>
  <head>  </head>
  <body>
    <font face="Arial"> Rajuk College </font>
  </html>
</body>
</html>
```

Sample MCQ					Ans.
1.	What is the HTML color code for black? a				
	a) #000000	b) #FFFFFF	c) #FF0000	d) #00FF00	1.a
2.	What is the HTML color code for white?				
	a) #FFFFFF	b) #000000	c) #FF0000	d) #00FF00	2.a
3.	Attributes of font tag are---- i) type ii) color iii) size				
	a) i & ii	b) i & iii	c) ii & iii	d) i, ii & iii	3.c
4.	Which is a valid html code?				
	a) 	b) 	c) <size="5" color="red" >	d) size="5" font color="red"	4.b
5.	What is the correct value for face attribute?				
	a) 7	b) red	c) Times New Roman	d) size	5.c

** tag:**

```
<!DOC Type HTML>
<html>
  <head>  </head>
  <body>
    This is a black text. But <span style="color:RED"> this is other text </span>
  </body>
</html>
```

Result:

This is a black text. But this is other text

Red color

Example:

I am fine. <div style="color:RED">This is a colorful line</div>

I am fine.
This is a colorful line

Red

How to set background color:

```
<!DOC Type HTML>
<html>
  <head> </head>
  <body style="background-color:red">
</body>
</html>
```

Another way to set background color:

```
<!DOC Type HTML>
<html>
  <head> </head>
  <body bgcolor="Yellow">
</body>
</html>
```

Sample MCQ

				Ans.
1.	Style attribute related with---			
	A) img	B) table	C) body	D) list
				1.c
2.	Which attribute used to change background color in html			
	a) br	b) style	c) img	d) list
				2.b
3.	The color of text can be changed by--- tag			
	i) Font			
	ii) Span			
	iii) Div			
	Which is correct			
	a) i & ii	b) i & iii	c) ii & iii	d) i, ii & iii
				3.d

List tag:

ul=unordered list

ol=ordered list

 tag in:

the tag in HTML is used to define an unordered list, where list items are displayed with bullets (•) by default.

Example1:

```

<!DOC Type HTML>
<html>
  <head> </head>
  <body>
    <ul>
      <li> Messi</li>
      <li> Ronaldo</li>
      <li> Neymar</li>
    </ul>
  </body>
</html>

```

- Messi
- Ronaldo
- Neymar

Example2:

```

<!DOC Type HTML>
<html>
  <head> </head>
  <body>
    <ul type="circle">
      <li> Messi</li>
      <li> Ronaldo</li>
      <li> Neymar</li>
    </ul>
  </body>
</html>

```

- Messi
- Ronaldo
- Neymar

Example3:

```

<!DOC Type HTML>
<html>
  <head> </head>
  <body>
    <ul type="square">
      <li> Messi</li>
      <li> Ronaldo</li>
      <li> Neymar</li>
    </ul>
  </body>
</html>

```

- Messi
- Ronaldo
- Neymar

Example 4:

```
<!DOC Type HTML>
<html>
  <head> </head>
  <body>
    <ul type="disc">
      <li> Messi</li>
      <li> Ronaldo</li>
      <li> Neymar</li>
    </ul>
  </body>
</html>
```

- Messi
- Ronaldo
- Neymar

** tag:**

The tag is used to create an ordered list, where items are displayed in a specific sequence using numbers or letters.

Example1:

```
<!DOC Type HTML>
<html>
  <head> </head>
  <body>
    <ol>
      <li> Messi</li>
      <li> Ronaldo</li>
      <li> Neymar</li>
    </ol>
  </body>
</html>
```

1. Messi
2. Ronaldo
3. Neymar

Example2:

```
<!DOC Type HTML>
<html>
  <head> </head>
  <body>
    <ol type="1">
      <li> Messi</li>
      <li> Ronaldo</li>
      <li> Neymar</li>
    </ol>
  </body>
```

1. Messi
2. Ronaldo
3. Neymar

Example3:

```
<!DOC Type HTML>
<html>
  <head> </head>
    <ol type="a">
      <li> Messi</li>
      <li> Ronaldo</li>
      <li> Neymar</li>
    </ol>
  <body>
</body>
```

a. Messi
b. Ronaldo
c. Neymar

Example 4:

```
<!DOC Type HTML>
<html>
  <head> </head>
    <body>
      <ol type="A">
        <li> Messi</li>
        <li> Ronaldo</li>
        <li> Neymar</li>
      </ol>
    </body>
</html>
```

A. Messi
B. Ronaldo
C. Neymar

Example 5:

```
<!DOC Type HTML>
<html>
  <head> </head>
  <body>
    <ol type="i">
      <li> Messi</li>
      <li> Ronaldo</li>
      <li> Neymar</li>
    </ol>
  </body>
```

i. Messi
ii. Ronaldo
iii. Neymar

Example 6:

```

<!DOC Type HTML>
<html>
  <head>  </head>
  <body>
    <ol type="I">
      <li> Messi</li>
      <li> Ronaldo</li>
      <li> Neymar</li>
    </ol>
  </body>
</html>

```

I. Messi
II. Ronaldo
III. Neymar

Example 7:

```

<!DOC Type HTML>
<html>
  <head>  </head>
  <body>
    <ol type="A" start="4">
      <li> Orange</li>
      <li> Mango </li>
      <li> Apple </li>
    </ol>
  </body>
</html>

```

D. Orange
E. Mango
F. Apple

Sample MCQ**Ans.**

- | | | | | | |
|---|------------|-------------|----------------|------------|-----|
| 1. Which is used to create list in html? | a) ul | b) th | c) tr | d) td | 1.a |
| 2. Which tag indicates each item of a list in HTML? | a) | b) | c) | d) <start> | 2.b |
| 3. Related tags of list in HTML are: | | | | | |
| i) ol | | | | | |
| ii) ul | | | | | |
| iii) li | | | | | |
| a) i & ii | b) i & iii | c) ii & iii | d) i, ii & iii | | 3.d |
| 4. What is the default type of an unordered list? | | | | | |
| a) square | b) circle | c) disc | d) round | | 4.c |

5. What is the default type of an ordered list?
 a) 1 b) a c) i d) A 5.a
6. The values used for ordered list are:
 i) 1
 ii) A and a
 iii) I and i
 which is correct?
 a) i & ii b) i & iii c) ii & iii d) i, ii & iii 6.d
7. The values used for unordered list are:
 i) disc
 ii) square
 iii) rectangle
 which is correct?
 a) i & ii b) i & iii c) ii & iii d) i, ii & iii 7.a
8. The output the below tag--

```
<ol Type= "A" start="3">
<li> Dhaka </li>
<li> Gazipur</li>
</ol>
```

 a) A. Dhaka 3. Gazipur b) C. Dhaka D.Gazipur c) 1. Dhaka C. Gazipur d) a. Dhaka c. Gazipur 8.b

Sample CQ	
Home page of ABC college has a list of different faculties in sequential form: Sc, B.St and humanities . Write down the html tag for home page of ABC college.	
○ Mahin ○ Nihan ○ Zarif ○ Rafin Write HTML code to the display the above figure.	
1. Month ○ January ○ February 2. Day ■ Sunday ■ Monday Write HTML code to the display the above figure.	
▪ Apple of a. Austalia b. Africa ▪ Orange ▪ Banana Write down the HTML code to show the webpage.	

** tag:**

The tag is used to display images on a webpage.

```
<!Doctype html>
<html>
<head> </head>
<body>
    
</body>
</html>
```

How to change size of an image:

```

```

Alt attribute:

The alt attribute provides alternative text if the image cannot be displayed.

Example:

```

```

Show image from a specific location:

Display an image named rail.jpg stored in the picture folder on the D: drive

```
<!Doctype html>
<html>
<head> </head>
<body>
    
</body>
</html>
```

Sample MCQ**Ans.**

- | | | | | |
|---|------------|-----------|----------|-----|
| 1. What does the HTML 'img' tag stand for? | | | | |
| a) Import | b) Imagine | m) Inline | d) Image | 1.d |
| 2. Which attribute used to show image? | | | | |
| a) source | b) src | c) jpg | d) img | 2.b |
| 3. Mr. Rupam tried to show an image on his home page. But the image was NOT displaying properly. the reasons can be ----- | | | | |
| i) The browser did NOT support | | | | |
| ii) The image name was NOT written properly | | | | |
| iii) did not mention the location of the image | | | | |
| Which is correct? | | | | |

- a) i & ii b) i & iii c) ii & iii d) i, ii & iii 3.d
4. What is full form of JPG/JPEG?
a) Joint Photographic Experts Group b) Joint Picture Experts Group c) Joy Photographic Export Group d) Joint Photoshop Extra Group 4.a
5. HTML support -----file.
i) .jpg
ii) .gif
iii) .psd
Which is correct?
a) i & ii b) i & iii c) ii & iii d) i, ii & iii 5.a
6. The size of an image can be changed by -----and ----- attribute.
A) length, weight B) width , height C) border, order D) volume, weight 6.b
7. Which tag is similar of image tag?
a) tr b) hr c) width d) font 7.b
8. 4th page of the website contains a picture name Pic.jpg and its size is 300 x 300 pixels. But the problem is; the HTML code for page No. 4 is :
a) < img src "Pic.jpg" height "300px" width "400px"> b) < img src = "Pic.jpg", length = "300" width = "300"> c) < image src = "Pic.jpg" height = "300" width = "300"> d) < img src = "Pic.jpg" height = "300px" width = "300px"> 8.d
9. The most important attribute with an img tag is -----
a) width b) alt c) src d) height 9.c
10. If webpage unable to show the picture which attribute will send a message?
a) alt b) post c) cancel d) hlt 10.a
- Tree is name of an image. It is saved in F drive under a folder named XYZ. The tag should be---
a) F:\ XYZ\Tree.jpg b) F:\ Tree.jpg\XYZ c) D:\ XYZ\Tree.jpg d) XYZ\Tree.jpg\F:\ 11.a

Sample CQ	
Joly developed a website. On her homepage, there is a picture of a tree. Write HTML code for Joly's home page	
 water Lily.JPEG	
Write html code to show the picture	
Mr. Arif developed a website. On his homepage, he wants to display a picture of his company. The image is 300 × 200 pixels and is stored in a folder named My Photograph on the E: drive of his computer. Write html code for the home page of Mr. Arif.	

Hyper link(anchor tag)

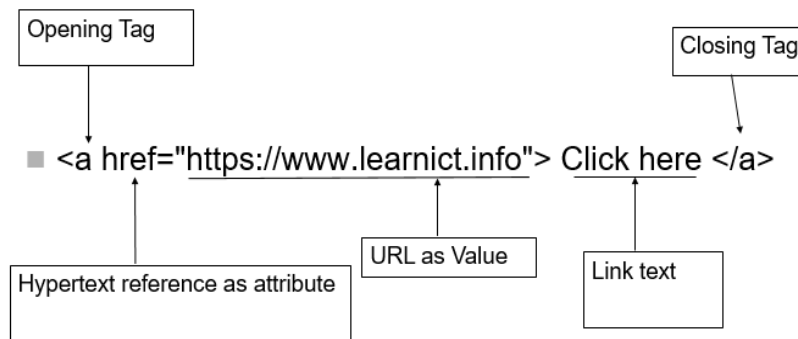
The <a> tag in HTML is used to create hyperlinks that connect one page to another, or link to files and external websites.

Example:

```
<a href="https://www.learnict.info"> Click here </a>
```



Different parts of hyperlink:



Internal hyperlink:

If both pages are within the same website, it is called an internal hyperlink.

Example:

If you want visit login.html page from home page then the html code will be:

```
<a href="login.html"> click here </a>
```

The importance for hyper link:

- Save time
- Easy to navigate
- Connects related information
- Provides quick access to external resources
- improve user experience
- Makes websites interactive
- Improves search engine ranking (SEO)
- Allows easy sharing of information

Sample MCQ

Sample MCQ					Ans.
1.	What is the attribute with anchor tag? A) href B) a C) url D) src				1.a
2.	Which is the correct example of a hyperlink in HTML? d) b)) c) <link href=a> d) <href a="URL">				2.b
3.	Hyper link includes— i) <a> ii) URL iii) hyper-reference Which of the following is correct? a) i & ii b) i & iii c) ii & iii d) i, ii & iii				3.d
4.	Which is the link text here? click here a) <a> b) href c) facebook.com d) click here				4.d
5.	Keya and Diya often go to each other's websites to collect the necessary information. It seems quite time-consuming for them. When they tell their IT teacher about the problem, he suggests them to use a special tag. Which tag is suggested by the IT teacher in the stem? a) href b) del c) a d) URL				5.c
6.	The advantages of hyper link are --- i) save time ii) easy navigation iii) visit a page or new website Which of the following is correct? b) i & iii b) i & iii b) i & iii b) i & iii				6.d
7.	Hyper link may connect--- i) two pages in a website ii) two sites in Internet iii) two fields of table Which of the following is correct? b) i & iii b) i & iii b) i & iii b) i & iii				7.a

Sample CQ

Shila visited a website, clicked on 'Our Board', and was taken to www.e-board.gov.bd

Q) Write HTML code for Shila's website.

The 'Home page' of Jaisa's website contains two links. One leads to facebook.com, and the other link leads to a page under the home page named 'details.html'.

Q) Write html code for Jaisa's home page

The website of a company has two links: Click 'Social Network' to open instagram.com. Another link goes to the product page of the company.

Q) Write HTML code for the above description.

Mr. Azim developed a website. In homepage there is a picture name **Bannar.jpg**. By Clicking on the image a visitor can go to visit Wikipedia.org.

Q) Write HTML code for Mr. Azim's homepage

Promita created 3 pages: Home.html, Admission.html and Result.html for developing a website. Then she make some connection system for going from home page to other pages.

Q) What are the importance of Promita's initiative

<table> tag:

Exercise:

Roll	Name
1	Dipu
2	Jashim

```
<table border="1">
<tr><th> Roll </th> <th> Name </th> </tr>
<tr> <td>1</td> <td> Dipu </td> </tr>
<tr><td>2</td> <td> Jashim </td> </tr>
</table>
```

```
<table border="1">
<caption> My table</caption>
<Tr> <th> Roll </th> <th> Name </th> </Tr>
<Tr> <td> 1 </td> <td> Dipu </td> </Tr>
<Tr> <td> 2 </td> <td> Jashim </td> </Tr>
</table>
```

My table

Roll	Name
1	Dipu
2	Jashim

Exercise:

ICT	
Sub	Obj

```
<table border="1">
<tr> <th colspan="2">ICT </th></tr>
<tr><td> Sub</td> <td>Obj</td> </tr>
</table>
```

Exercise:

```
<table border="3">
<tr> <th rowspan="3"> College </th> <td>Sc </td> </tr>
<tr> <td>B.St </td></tr>
<tr> <td>Hum</td></tr>
</table>
```

College	Sc
	B.St
	Hum

Exercise:

```
<table border="1" cellpadding="20">
<tr><th>Roll</th><th>Name</th></tr>
<tr><td> 101</td><td> ABC</td></tr>
<tr><td>206</td><td>XY</td></tr>
</table>
```

Roll	Name
101	ABC
206	XY

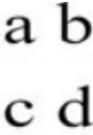
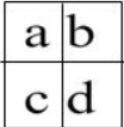
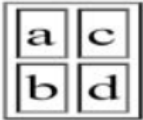
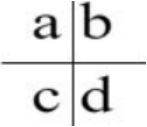
Exercise:

```
<table border="1" cellspacing="10">
<tr><th>Roll</th><th>Name</th></tr>
<tr><td> 101</td><td> ABC</td></tr>
<tr><td>206</td><td>XY</td></tr>
</table>
```

Roll	Name
101	ABC
206	XY

Sample MCQ

- | | Ans. |
|--|------|
| 1. Which is the most important attribute of a table tag?
a) border b) caption c) cellpadding d) cellspacing 1.a | |
| 2. Which element is used to define a cell in an HTML table?
a) caption b) th c) header d) td 2.d | |
| 3. Related terms of tables are ---
i) td
ii) tr
iii) strike
Which is correct?
a) i & ii b) i & iii c) ii & iii d) i, ii & iii 3.a | |

4. Which tag is used for creating the row of a table?
 a) td b) tr c) th d) cell 4.b
5. Cellspacing is attribute used with ---tag
 a) td b) table c) th d) tr 5.b
6. Which attribute joins multiple row of a table?
 a) colspan b) Rowadd c) Rowsapn d) Rowjoin 6.c
7. Which means table heading ?
 a) td b) tr c) th d) h1 7.c
8. What is the output of the following tag?
`<table>`
`<tr><td>a</td><td>b</td></tr>`
`<tr><td> c</td><td> d</td></tr>`
`</table>`
 a)  b)  c)  d)  8.a
9. Which one is connected to table tag?
 a) img b) del c) caption d) font 9.c

Sample CQ

Roll	Subject
201	Bangla
202	English
203	ICT

Write html code to display the above table if opened by chrome.

HSC -2019

ICT	
Subject	Board

Write HTML code so that clicking the "Board" link opens www.board.edu.bd to view ICT marks.

A		C	D
E	BF	G	H
I	J	KL	

Write html code to display the above table

```
<html>
<head><title>ICT</title></head>
<h3> COLLEGE RESULT</h3>
<table border="1">
<tr>
<th>Roll</th> <th>Name</th> <th>Result</th>
</tr>
<tr>
<td>501</td><td>Sumaiya</td>
<td> <a href="Exam Result.html"> MY test Result</a></td>
</tr>
</table>
</html>
```

What will be the output of the above html code?

A	B	
C	D	
E	F	

Write html code with inserting a picture in the blank cell

Test result		
Roll	Name	Result
101	Reza	A+
102	Arif	B+

Write HTML code so that clicking on image.jpg opens www.moedu.gov.bd

Student Name	Compulsory			Optional
Harry Potter	Bangla	English	ICT	Physics
				Math
				Biology

- c) Write html code to display the above table
d) Write html code to display the table by converting all rows into columns and vice versa

More Sample CQ

A school named “Alor Shopan” was planning to develop a website. On its homepage, a picture named “Campus.jpg” will appear, and a list of sections such as IT, Physical Science, and Biological Science will be arranged. There will also be a ‘Notice Board’ linked on the homepage.

The school authority invited three IT specialists. The IT specialists suggested two solutions. The first one was cheaper but had some difficulties with regular updates. The second solution was more costly but easy to update on a regular basis.

C) Write HTML code to create the homepage.

D) Which solution is better for Alor Shopan?

Mr. Z was demonstrating HTML coding in a programming class. He added a picture from the C drive under a folder named “Picture.” The name of the image was “Logo.jpg” and its size was 300 × 300 pixels. Then he asked his students to write an HTML code that would allow the image to be clickable and redirect to xiadmissionboard.edu.bd. After that, he wrote:

C) What HTML code was written by the students? Explain its attributes.

D) According to the stem, what steps should be taken to make people interested in the institute?

A Website has been designed for "Chandana Model College" using HTML only. There is a picture named ict.jpg of size 200x300 Pix in the home page of the site. There have a link of notice page named notice.html under the picture. The text "Welcome to Chandana Model College" is displayed on the top of the picture in blue color. There is no arrangement to give visitors opinion in that website.

C) Write down the HTML, Code to develop the Home page as mentioned in stem.

D) Explain necessary steps taken so that visitors can give their opinion in website.

XYZ College, Dhaka.

Available Honor's subject:

1. Bangla
2. English
3. Mathematics
4. Accounting

c) Write the HTML code for showing the stem in your college's home page.

d) Write HTML code for taking the list of Honor's subject and make a table. There will be a table with headline serial No and subject name.

Mr. Azim developed a website. In homepage there is a picture name **pic.png**. By Clicking there a visitor can go to visit **bing.com** the size of the picture is 200 x 200 and it is kept in a folder named ABC in

D drive of hard disk.
Write HTML code for Mr. Azim's homepage

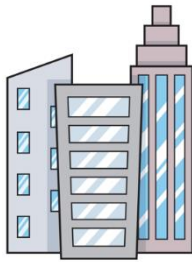


Figure-1

A) Car
B) Bus
C) Truck
Prev

Figure-2

c) Write HTML code to display Figure-1.

d) Write HTML code to display Figure-2, where clicking "Prev" will open Figure-1.

Website design and development:

Some key points to keep in mind while designing a website:

- The design should look attractive to users.
- Users should be able to find different features easily.
- The features should be simple and usable.
- The website should load quickly on all devices.
- Security and privacy of users should be ensured.

The design of a website depends on its purpose:

For example, consider a website where users can ask and answer programming-related questions. Before designing, we must first decide on the key features. These may include:

- User registration and login system.
- Ability to post questions.
- Ability to post answers to existing questions.
- Option for the question owner to select the best answer.
- Voting system for helpful answers.
- A points or reputation system to reward good questions and answers.

Next, we decide which pages the website will require.

For each page, we design a layout that defines the placement of elements. At the initial stage, this layout can be sketched as a *wireframe* using pen and paper. Later, graphic design tools such as

Adobe Illustrator
GIMP.

Two Parts of Building a Website

- 1) Front-End Development
- 2) Back-End Development

1) Front-End Development (What you see)

- Uses HTML and CSS to build web pages based on design.

- Called *front-end development*.
- Requires knowledge of more tools beyond HTML & CSS.

Back-End Development (The hidden brain)

- Involves programming for features and database connection.
- Called *back-end development*.
- Developer with both skills = *full-stack developer*.

Full-Stack Developer: A developer who knows how to do **both** front-end and back-end development.

Important Things during Development:

- **Design Can Change:** Requirements often change, so you should write code that is easy to modify later.
- **Testing is a Must:** You must constantly test the website and fix bugs (**debugging**) to make sure everything works perfectly.

Publishing a Website (How to Launch It Online)

Steps to publish a website:

1. Register a Domain
2. Host the Website.
3. Connect Domain and Server

1) Register a Domain (Naming Your Website)

Problem: IP addresses (e.g., 192.0.2.44) are hard to remember.

Solution: Use a Domain Name (e.g., wikipedia.org).

DNS(Domain Name System): It is a system that translates domain names into their corresponding IP addresses so browsers can find websites.

2) Web Server(hosts the website)

To make your site available to everyone, your files must live on a special computer called a web server.

A web server must: Have a Public IP Address (its unique location on the internet).

Be on 24/7.

Since personal computers are not practical for this, we use a Hosting Service.

Hosting: You pay a company to rent space on their server and upload your files there.

Example: neapcheap.com, godaddy.com

How to get more visitors (make people interested) to a website

- Make it user-friendly
- Do SEO (Search Engine Optimization)
- Share on social media
- Post useful and fresh content
- Use catchy titles and headings
- Make the website mobile-friendly
- Use fast loading pages

- Add links to other websites (and get backlinks)
- Advertise online (Google, Facebook Ads)
- Use keywords people search for
- Offer something free (guide, tool, etc.)
- Engage with visitors(chat, comments)

Ans.

Sample MCQ

1. What is step after design and development of an website?
 A) collection of data B) wireframe C) publishing D) system analyze 1.b
2. Which programming languages are commonly used for front-end development?
 a) Python and Ruby b) PHP and SQL c) HTML and CSS d) Java and C++ 2.c
3. Steps of website publishing are:
 i. Register a Domain
 ii. Host the Website
 iii. Connect Domain and Server
 Which is correct?
 a) i & ii b) i & iii c) ii & iii d) i, ii & iii 3.d
4. Who develop both front end and back of website?
 a) Front end developer b) Back end developer c) Middle point developer d) Full stack developer 4.d
5. What is the purpose of a public IP address in website hosting? To improve website security
 a) To restrict access to the website b) To identify the location of the server c) To manage web performance d) d) To improve website security 5.b
6. Debugging means—
 A) create B) draw C) correction D) release 6.c
7. What is the process of making a website accessible to all internet users?
 a) Publishing b) Design c) Development d) Maintenance 7.a
8. A web server must---
 i. have a public IP
 ii. Run 24 hours
 iii. Connect to the internet
 Which is correct?
 a) i & ii b) i & iii c) ii & iii d) i, ii & iii 8.d
9. Which one can identify a computer uniquely?
 A) Public IP B) Private IP C) host PC D) browser 9.a
10. Task of back end of website--
 i. Connect database
 ii. Sending email
 iii. Implement features of websites

- 11 a) i & ii b) i & iii c) ii & iii d) i, ii & iii 10.d
- Publishing includes ---
- A) browsing and upload B) wireframe and design C) domain and hosting D) HTML and tags 11.c

Sample CQ	
Shuvo received some training on web page development. He created a webpage with his picture and information, and he could run it from his own computer. He wanted to improve his site and name it “ shuvo.net. ” Knowing this, his friend Rahul said, “ <i>I want to visit your webpage tonight.</i> ” In reply, Shuvo said, “ <i>Not now. I still need to do some publishing work.</i> ”	
Q) Explain Shuvo’s comment.	
The picture of the administrative building of the ‘X’ institute shown in the home page of it’s website. If the website published in Internet , awarded people of the world will be informed about the institute.	
Q) According to stem what steps should be taken to make people interested about the institute	
A company named Delta wants to develop a website where they will only introduce themselves. They appointed a web developer for that work. The web developer told the authority that to publish the website, they need a domain name and hosting.	
Q) What is the difference between domain and hosting, as expressed in the stem?	
A website can be created and run on a personal computer, but for others to access it worldwide, it must be placed online. To make this possible, the site’s files need to be stored on a special server connected to the Internet. This allows people from anywhere to open the site using its web address. In this way, a simple local webpage becomes available to everyone.	
Q) What activities reflected in the stem?	